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Progress Report

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Introduction

Global Navigation Satellite Systems (GNSS) provide location and timing information to the users worldwide and have become a crucial part for positioning and navigation in the modern era, especially GNSS-integrated smartphones are on the rise, enabling mobile positioning and location-based services. However, for standard GNSS, there are various error sources such as satellite clock error, ionospheric and tropospheric delays, multipath effects, and receiver noise. To obtain a high accuracy position, these errors must be corrected or mitigated.

Differential GNSS (DGNSS) is one of the techniques used for improving the positioning accuracy by applying the corrections sent from reference stations with known coordinates, ultimately correcting some errors such as satellite clock errors, and mitigating ionospheric and tropospheric delays. Since some GNSS errors are spatially correlated, it is assumed that close stations have the exact same errors, therefore the errors can be cancelled. Traditionally, DGNSS is often implemented in GNSS receivers to increase their accuracies. With the rise of smartphone usage and the ability to receive raw GNSS messages, it is now feasible and increasingly common to implement DGNSS corrections on mobile devices.

In Hong Kong, the Lands Department (LandsD) provides Satellite Positioning Reference Station Network (SatRef) service, which sends data stream containing real-time correctional data to the users via SatRef Networked Transport of RTCM via Internet Protocol (NTRIP) server (LandsD, 2025). To integrate DGNSS into mobile devices in Hong Kong, SatRef is often used since the mobile devices are spatially close to the SatRef stations and public access to well-established positioning infrastructure.

This project develops and evaluates a mobile based DGNSS algorithm using raw GNSS measurements and SatRef correction data, analyzes the achieved accuracy, and identifies the key limitations and improvements.

Project Objectives

- Develop algorithm

This project aims to implement the full GNSS positioning workflow into a mobile app using Kotlin. This includes decoding GNSS navigation messages (satellite ephemerides data) and RTCM correction data, handling GNSS measurements and time systems, computing satellite positions from ephemerides, and estimating the receiver position using least squares adjustment. Since Android APIs only cover limited abstractions over navigation, observation data, most of the received binary data needed to be decoded manually.

- Analyze accuracy

  This project also aims to evaluate the positioning accuracy of the resolved location. This involves surveying geodetic control stations with known coordinates, and comparing the positions from standalone GNSS and DGNSS-corrected against the known points, thus the horizontal and vertical accuracy can be obtained for analysis.

The analysis can provide insights for identifying the strengths and limitations of DGNSS on mobile device.

- Recommend improvements

  With the survey results collected and the limitations identified in the analysis, this project also aims to propose technical and practice improvements. These include enhancements to error correction models, user interface design, change in algorithm and extensibility of the DGNSS positioning software.

Literature Review

1. GNSS

1a. Introduction and principals of GNSS

GNSS serves as space-based tools for finding one's position across the globe. Major satellite system constellation examples include the GPS from the United States, GLONASS from Russia, Galileo from Europe, and BeiDou from China. According to Kaplan and Hegarty (2017), a GNSS system consists of three components, space segment, control segment (CS), and user segment. Space segment refers to the satellite constellation, which is a collection of satellites that orbit the earth and return ranging signals and data messages, for instance satellite ephemeris data back to the users. The control segment maintains the satellites, for instance the satellite health, battery levels, updates the satellite's clock and ephemeris, overall ensuring the satellites can properly function. Lastly the user segment refers to the GPS receivers, which receive the signals transmitted from the satellites and perform the computations, ultimately resolving their own positions.

Positioning relies on measuring the time a signal takes to travel from the satellite to a receiver. Since the signals travel with EM waves, by multiplying the travel time by the speed of light, the receiver can resolve the distance from each satellite. The distance calculated between the satellite and the receiver is the pseudorange, which is an approximate distance and contain errors that need to be corrected, such as atmospheric delays and clock errors. The ephemeris data can provide the 3-D location relative to the earth's center, and at least four satellites is required to solve for the position of the receiver by solving the least squares solution.

1b. Positioning techniques

The most common and widespread positioning method is single point positioning, or SPP (Rosalie et al., 2017), since post-processing is not required, only uses one receiver and at least 4 satellites, and usually used for resolving one's position instantly. While SPP is operationally simpler, and does not depend on other structures such as a local base station or network connection, SPP usually has more errors and might only be accurate to meter-level. Therefore, it is usually used in coarse positioning where meter level accuracy is acceptable (Santerre et al., 2014).

One of the alternative positioning techniques is precise point positioning (PPP), and can often achieve higher accuracy up to sub-decimeter level. However, according to Afifi and El-Rabbany (2016), one major weakness in PPP is that it takes relatively long time to converge the solution. Due to the accuracy it can provide, many precise satellite equipment has incorporated with PPP positioning technique, however, it is likely infeasible for mobile applications, since the time to resolve ones position is its major drawback.

On the other hand, real time kinematic (RTK) is relatively fast and uses a base station to correct the errors, and can often achieve decimeter level accuracy (Liu et al., 2022). However, the setup is generally more complex and handling carrier phase data. Furthermore, since carrier phase data is precise yet very sensitive, it is generally more susceptible to multipath errors caused by building blockage, and non-line-of-sight (NLOS) can significantly degrade the signal quality and thus the positional accuracy could be affected (Liu et al., 2022).

Last but not least, differential GNSS (DGNSS) is also an alternative positioning technique that is based on SPP, but also uses the correctional data sent from the base station to the rover station or the receiver, and can often achieve meter-level accuracy (Kaplan & Hegarty, 2017).

In this project, SPP and DGNSS were used. Therefore, we focus on the position calculation techniques for SPP for the basic position before adjustments.

Before resolving the satellite's position, its orbit needs to be known, and the six Keplerian elements define the satellite orbit (Liu, 2025). Among the elements, except true anomaly that needs to be calculated, all other values are sent from the satellites in the navigation data.

a	Semi-major axis	define size and shape of orbit
e	Eccentricity	
Ω	Right ascension of ascending node	Orientation of the orbital plane in the inertial system
ω	Argument of perigee	
i	Inclination	
θ	True anomaly	Angle between satellites and the perigee. It denotes the position of the satellite in the orbital plane.

According to Liu (2025), after retrieving the ephemeris data containing semi-major axis a and argument of perigee ω, the next steps are to calculate the mean motion (n), mean anomaly (M), eccentric anomaly (E), true anomaly (θ), orbit radius (r), and GPS satellite position (x, y, z) in the orbital plane. They can be calculated by the following equations respectively.

Mean motion (n)	$n = \frac{\sqrt{u}}{\left(\sqrt{a}\right)^3} + \delta n$
Mean anomaly (M)	$M_k = M_0 + t_k(n_0 + \delta n)$

Eccentric anomaly (E)	$E_k = M_k + e \sin(E_k)$
True anomaly (θ)	$\theta_k = 2 * atan\left(\frac{sqrt(1 + e)}{sqrt(1 - e)} tan\left(\frac{E}{2}\right)\right)$
Orbit radius of the satellite position (r)	$r = a(1 - e \cos E_k) + \delta r_k$ $\delta r_k = C_{rs} \sin(2\varphi_k) + C_{ic} \cos(2\varphi_k)$ $r_c = r_k + \delta r_k$
GPS Position in orbital plane	$x_0 = r_c \cos(\varphi_c)$ $y_0 = r_c \sin(\varphi_c)$
Uncorrected GPS position in ECEF frame	$x = x_0 \cos(\Omega) - y_0 \cos(i) \sin(\Omega)$ $y = x_0 \sin(\Omega) + y_0 \cos(i) \cos(\Omega)$ $z = y_0 \sin(i)$

At this point, the GPS position in Earth-centered, Earth-fixed (ECEF) coordinate system can be calculated, but the argument of latitude, inclination, orbital radius can be corrected. The longitude of ascending node can also take into account of the Sagnac effect, which must be corrected before resolving the receiver's location.

Corrected inclination (i)	$i_k = i_0 + \delta i + \delta i_k * t_k$ $\delta i_k = C_{is} \sin(2\varphi_k) + C_{ic} \cos(2\varphi_k)$ $i_c = i_k + \delta i_k$
Corrected orbital radius (r)	$r = a(1 - e \cos E_k) + \delta r_k$ $\delta r_k = C_{rs} \sin(2\varphi_k) + C_{ic} \cos(2\varphi_k)$ $r_c = r_k + \delta r_k$
Corrected Argument of latitude	$\delta\varphi_k = C_{us} \sin(2\varphi_k) + C_{uc} \cos(2\varphi_k)$ $\varphi_c = \varphi_k + \delta\varphi_k$
Longitude of ascending node, corrected for Sagnac effect	$\Omega_k = \Omega_0 + t_k(\delta\Omega - \delta\Omega_e) - \delta\Omega_e * t_0$
Corrected GPS position in ECEF frame	$x = x_0 \cos(\Omega_k) - y_0 \cos(i_c) \sin(\Omega_k)$ $y = x_0 \sin(\Omega_k) + y_0 \cos(i_c) \cos(\Omega_k)$ $z = y_0 \sin(i_c)$

The equations for least square adjustments can be described as below:

$$B = \begin{bmatrix} \frac{x_0 - x}{\rho} & \frac{y_0 - y}{\rho} & \frac{z_0 - z}{\rho} & 1 \\ \dots & \dots & \dots & \dots \end{bmatrix}$$

$$f = \begin{bmatrix} c1 - \rho + \mathrm{dT} \\ \dots \end{bmatrix}$$

$$adjustments = (B^T * P * B)^{-1} * B^T * P * f$$

If we assumed each satellite had identical weighting, then P would be an identity matrix and is omitted from the calculation.

$$P \approx \rho + \mathrm{c}(\mathrm{dt} - \mathrm{dT}) + I$$

Therefore, adjustments can be approximated with the following equation:

$$adjustments = (B^T * P * B)^{-1} * B^T * P * f \approx (B^T * B)^{-1} * B^T * f$$

1c. Position correction techniques

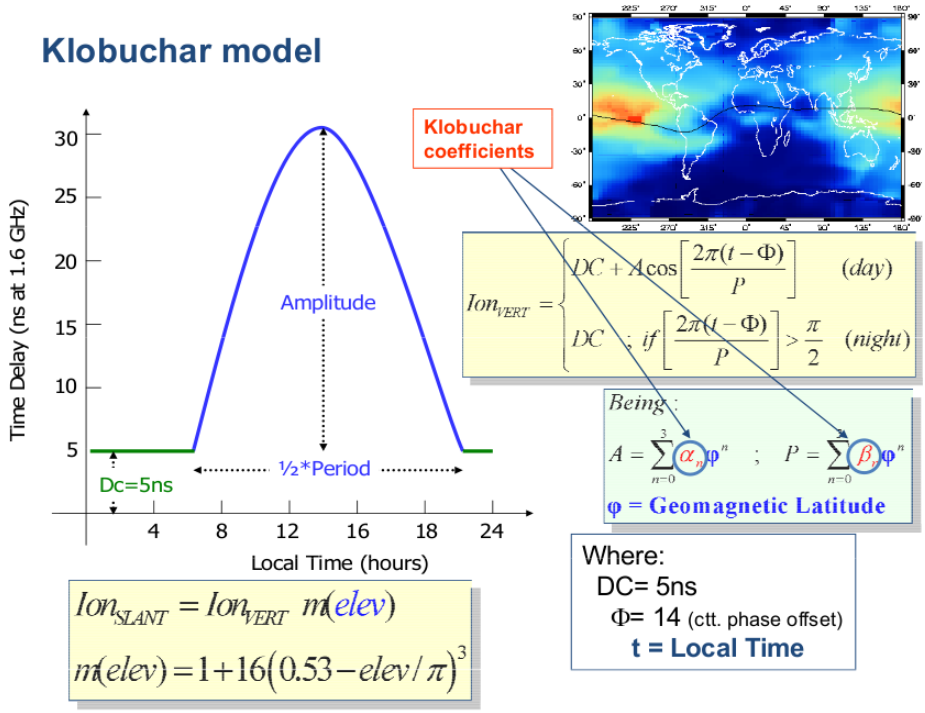
There are many error sources in GNSS positioning, such as satellite clock error, receiver clock error, ionospheric and tropospheric error, multipath error, and noise (Liu, 2025). The errors can be described as the following equations:

$$\Phi = \rho + \mathrm{d}\rho + \mathrm{c}(\mathrm{dt} - \mathrm{dT}) - \mathrm{I} + \mathrm{T} + \mathrm{e}_{\mathrm{m}\varphi} + \lambda \mathrm{N} + \mathrm{n}_{\varphi}$$

$$\mathrm{P} = \rho + \mathrm{d}\rho + \mathrm{c}(\mathrm{dt} - \mathrm{dT}) + \mathrm{I} + \mathrm{T} + \mathrm{e}_{\mathrm{mP}} + \mathrm{n}_{\mathrm{P}}$$

Φ: carrier phase range in m	T: tropospheric error
ρ: pseudorange code range in m	N: ambiguity in carrier phase range (cycles)
$d\rho$: orbit error	$e_{m\varphi}$: multipath error in carrier phase range
dt: receiver clock error	e_{mP}: multipath error in pseudorange
dT: satellite clock error	n_{φ}: noise in carrier phase range
I: ionospheric error	n_P: noise in pseudorange

According to Liu (2025), the ionosphere has daily variations in electron density, solar activity, and solar cycle. To mitigate ionospheric delay, we can apply the Klobuchar model, as it is adopted by GPS.



2. DGNSS

2a. Introduction about DGNSS

According to Bonnor (2011), DGNSS was one of the first positioning techniques to be widely adopted, commonly used in maritime and aviation navigation. According to Kaplan and Hegarty (2017), differential GPS (DGPS) extends the functionality of a GPS receiver that uses standard positioning service (SPS), by having at least one reference station with known coordinates. The reference station can compare the measured pseudorange against the theoretical ranges to derive a correction value for each measured satellite, and sent to the end user data that may contain corrections to the receiver's pseudorange measurements, clock, and ephemeris data. Although there are more accurate alternatives such as the RTK, which can provide superior precision, DGNSS remains highly relevant and feasible to implement for mobile applications, due to its implementation being simpler than RTK.

2b. Principals of DGNSS in position correction

The mathematical principals of DGNSS is its nature of cancelling or mitigating errors from two sets of pseudorange equations. According to Liu (2025), after forming the equation for the rover station and receiving the pseudorange correction (PRC) message, the corrected pseudorange can be expressed as follows:

$$
\begin{aligned}
P^p_{m_corr} &= \rho^p_m + \cancel{orb^p} + c(dt_m - \cancel{dT^p}) + I^p_m + T^p_m + M^p_m + n_m \\
&\quad - \cancel{orb^p} - c(dt_b - \cancel{dT^p}) - I^p_b - T^p_b - M^p_b - n_b \\
&= \rho^p_m + c(dt_m - dt_b) + (I^p_m - I^p_b) + (T^p_m - T^p_b) \\
&\quad + (M^p_m - M^p_b) + (n_m - n_b)
\end{aligned}
$$

By the first law of Geography, it is assumed that the ionospheric and tropospheric delays are spatially correlated, in other words, the closer the rover and the base station is, the lower the delay error. To progress forward, we simply assume the errors at the base and mobile stations are the same, thus eliminating the terms in the equation.

$$
\begin{aligned}
P^p_{m_corr} &= \rho^p_m + orb^p + c(dt_m - dT^p) + I^p_m + T^p_m + M^p_m + n_m \\
&\quad - orb^p - c(dt_b - dT^p) - I^p_b - T^p_b - M^p_b - n_b \\
&= \rho^p_m + c(dt_m - dt_b) + \cancel{(I^p_m - I^p_b)} + \cancel{(T^p_m - T^p_b)} \\
&\quad + (M^p_m - M^p_b) + (n_m - n_b)
\end{aligned}
$$

The corrected pseudorange at the mobile station can be expressed as the following equation, with difference in M and n being multipath and noise respectively.

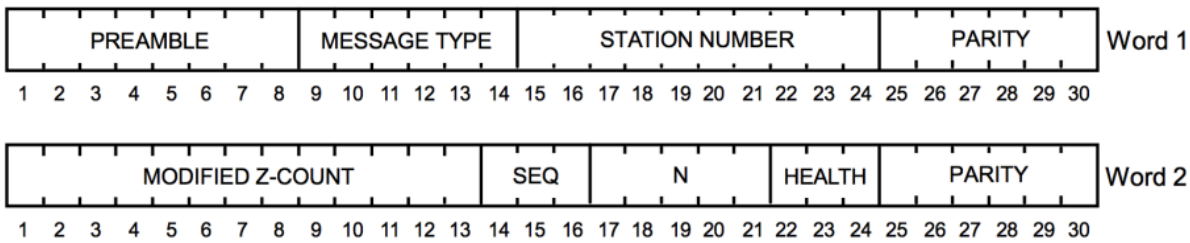
$$
P^p_{m_corr} = \rho^p_m + c(dt_m - dt_b) + (M^p_m - M^p_b) + (n_m - n_b)
$$

Since the errors vary across sites and applying error models for general case would be infeasible, they cannot be easily cancelled by making the differential equation. They can be classified as random errors. Other than such errors, the only error left in the equation is the receiver clock error, which can be estimated using least squares adjustments previously mentioned.

2c. Introduction about RTCM data

Radio Technical Commission for Maritime Services (RTCM) is a common data format used for maritime radio communications and navigation. For example, SatRef service sends the correction data to the users using RTCM, with version 3 for RTK and version 2 for DGNSS (LandsD, 2026). According to Lin (2006), RTCM v2 messages consists of 30-bit words. The first two words are the headers while the rest is the body message.

Below shows the two-word header structure (RTCM, 2001).



The preamble marks the start of the header and is always 0x66 (01100110 in binary), the next 6 bits are the message type. The next 10 bits is the station number. In the end of each 30-bit

word is the parity bits. The hamming code (32, 26) is applied to the first 24 bits, and the parity bits should match the last 6 bits.

The following table shows the meaning of the RTCM message types (Liu, 2025).

Message type no.	Status according to [1]	Name
1	**Fixed**	**Differential GPS Corrections**
2	Fixed	Delta Differential GPS Corrections
3	**Fixed**	**GPS Reference Station Parameters**
4	Tentative	Reference Station Datum
5	**Fixed**	**GPS Constellation Health**
6	**Fixed**	**GPS Null Frame**
7	**Fixed**	**DGPS Radiobeacon Almanac**
8	Tentative	Pseudolite Almanac
9	**Fixed**	**GPS Partial Correction Set**
10	Reserved	P-Code Differential Corrections
11	Reserved	C/A-Code L1, L2 Delta Corrections
12	Reserved	Pseudolite (Pseudo-Satellite) Station Parameters
13	Tentative	Ground Transmitter Parameters
14	Tentative	GPS Time of Week
15	Tentative	Ionospheric Delay Message
16	**Fixed**	**GPS Special Message**

In the second word, the first 13 bits is the Z count, which is the number of 0.6-seconds in an hour, which gives the reference time information of the base station to the receiver. The value counts from 0 to 6000 and wraps back at zero. The length of frame indicates the number of words of the message, so the receiver can know when to stop when decoding the message. The station health can indicate reliability of the data. For example, 0 indicates reliable, 6 indicates transmission is unmonitored, and 7 indicates the station is not functioning properly.

It should be noted that the sign of the data bits is determined by the parity bits of the previous word (rtcm-ntrip.org, 2008), if the top 2 bits are not 0x00 (00), it indicates the 24 bits of data are bit-inverted. This decoding process makes the next word reliant on the state of the previous word, which can likely reduce decoding errors.

```
private fun retainOrInvertData(word: Int): Int {
    // The top 2 bits are not 0x00, indicating an inversion
    if ((word and 0x40000000) != 0) {
        // Top 2 bits xor 0, first 24 bits (data) xor 1, last 6 bits (parity) xor 0
        // Xor 1 flips bits, xor 0 retains bits
        return word xor 0x3FFFFFC0
    }
    // Retains all bits
    return word
}
```

Implementation in the Project application, reading subsequent words as 32-bit integer, handling the case where the raw NTRIP data might invert the 24-bit data.

2d. DGNSS correction techniques

DGNSS uses a relatively simple correction technique, by applying the pseudorange correction (PRC) to the uncorrected pseudorange, then perform the least squares adjustment to obtain the adjusted receiver position. The equations can be expressed as the following.

$$PRC(t)_{base} = PRC(t0)_{base} + RRC(t0)_{base} * (t-t0)$$

Pseudorange correction (PRC) and range rate correction (RRC) are broadcasted from the base station to the receiver, and the magnitude of PRC is much larger than RRC, t is the target time, when the receiver receives the correction data, and t0 is the reference time when the base station receives the satellites, since there is a transmission latency between the base station and the rover, RRC accounts for the latency and applies the correction to the final PRC.

However, sometimes the receiver might receive multiple correction data from different base stations for the same satellite, and we might select the station closest to the receiver for single base station DGNSS. This would be feasible for long-range DGNSS as the base stations can be far apart, and selecting a single closest one is sufficient. However, in Hong Kong multiple stations would be relatively close to the receiver, in this case we can select the correction data based on the metric. For example, RTKLIB implements a heuristic variance model combining user differential range error (UDRE), station health (k), and latency (Δt) to form a variance of fast correction.

```
*var=varfcorr(p->fcorr.udre)+degfcorr(p->fcorr.ai)*t*t/2.0;
```

$$variance = udre^2 + k * \frac{\Delta t^2}{2}$$

$$weight = 1/variance$$

Then the correction data with the highest weight can be selected, often indicating high station health, low URDE, and low latency.

2e. Advantages of using DGNSS for mobile devices

For mobile devices, DGNSS offers a balance between accuracy and computational complexity. Since most mobile devices usually use low-cost antennas, and some might not be able to receive the carrier phase data, which is required for RTK. In contrast, DGNSS can operate as long as there is internet to connect to the NTRIP client and receive RTCM data, and the device is equipped with GNSS antenna to receive satellite ephemerides and pseudorange measurements, giving DGNSS a wider smartphone user base. According to Liu (2025), the benefits of using DGPS correction through the Internet is that the range is not

limited to line-of-sight distance, not affected by atmospheric attenuation, easy to set up, and low costs.

3. SatRef

3a. Introduction about SatRef service

The Satellite Positioning Reference Station Network (SatRef) is a geodetic infrastructure operated and managed by the Lands Department (LandsD) of the Hong Kong SAR Government. This network is equipped with GNSS receivers and monitors the GNSS signals continuously. SatRef is designed to send the correction data in RTCM format to the user through NTRIP. These stations ensures that the general public of Hong Kong can have access to reliable correction data after registration, and its service is free of charge.

3b. Advantages of using SatRef service for mobile devices

Firstly, using SatRef tends to be more accurate than using other DGNSS base stations. By using the SatRef network for mobile DGNSS, the distance between the rover stations and the base stations is reduced and thus the errors can be minimized. In DGNSS positioning, using local base stations is much more preferred than using inter-city ones, due to the ionospheric and tropospheric delays are spatially correlated. By minimizing the delays, the location accuracy can be further improved.

Secondly, since the network is maintained by Hong Kong governmental department, the data quality tends to be high, and the data tends to be more reliable.

Lastly, testing with SatRef is very accessible. Since LandsD provides the data free of charge and only a prompt registration is required, this makes location based app developers, academic researchers, and navigation users easily access the data from SatRef.

Methodology

The following is the main function for processing the received DGNSS signals.

```
while (isRunning) {
    bytesRead = inputStream?.read(buffer) ?: break
    // Immediately record the receiver time
    val recvTime = GTime.fromUnixNow()

    // Append to messageBuffer and parse RTCM messages
    val message = buffer.copyOf(bytesRead)

    // Try to extract and parse the next RTCM message
    val rtcm = RtcmDecoder.decodeRtcm(message) ?: continue
    val stationId = rtcm.stationId
    Log.d( tag: "GNSS",  msg: "Received DGNSS message type ${rtcm.msgType} from station $stationId!")
    rtcm.recvTime = recvTime
    pushDgnssMessage(rtcm)
    stationRtcmMap[stationId] = rtcm
}
```

The input stream can yield a byte array once an RTCM message is received, or else it blocks the current thread. The stream is from a socket connected to www.geodetic.gov.hk with port 2101 and mount point DGPS (LandsD, 2026).

To account for multiple base stations, a map with station ID as the keys is used to store the RTCM data.

Below shows the main function for adjusting the pseudoranges using DGNSS data. We first search the station RTCM map using the PRN of the pseudoranges. This ensures that every satellite has been checked to use DGNSS. Then the weight of the satellite is calculated. If there were no correction data of this PRN in the map, skip and move to the next PRN.

```
private fun adjustPseudorange(obsDataList: List<ObsDataWithRange>, stationRtcmMap: Map<Int, Rtcm>) {
    for (obsData in obsDataList) {
        val prn = obsData.inner.prn

        var bestRtcm: Rtcm? = null
        var bestWeight = Double.NEGATIVE_INFINITY
        for (rtcm in stationRtcmMap.values) {
            val weight = rtcm.weightOfSat(prn) ?: continue
            if (weight > bestWeight) {  // w = 1/variance, the higher the better
                bestRtcm = rtcm
                bestWeight = weight
            }
        }

        // Skip if no PRN or not type 1
        val rtcm = bestRtcm ?: continue
        val dgps = rtcm.dgps[prn - 1] ?: continue
        val age = rtcm.recvTime.asGpsTimeSecs() - dgps.t0.asGpsTimeSecs()
        val correctedPseudorange = obsData.pseudorange + dgps.prc + dgps.rrc * age
        obsData.pseudorange = correctedPseudorange
    }
}
```

With the previously measured receiving time, and the decoded reference time, PRC, RRC, the corrected pseudorange can be calculated before applying the least squares adjustment.

Preliminary Results

Five Hong Kong geodetic benchmarks and traverse control stations were surveyed and below are the results of standalone single point positioning without any correction techniques. The latitudes and longitudes are in WGS84, while eastings and northings are in HK1980.

Station ID	Measured Latitude	Measured Longitude	Measured Height
231205	22.386662482	114.19981660	71.3854
231207	22.380589774	114.19370649	-1.2816
8100.001	22.414759403	114.21240646	-64.9467
8056.001	22.447881194	114.17155947	68.4529

8032.005	22.447203914	114.17206789	70.3231

The following data are the coordinates with DGNSS applied.

Station ID	Measured Latitude	Measured Longitude	Measured Height
231205	22.386593422	114.19981591	71.3854
231207	22.380545857	114.19368590	-1.2816
8100.001	22.414755947	114.21238099	-64.9467
8056.001	22.447926786	114.17157938	68.4529
8032.005	22.447173580	114.17204339	70.3231

The following data is from Search of Geodetic Survey Control Station, LandsD.

Station ID	Northing	Easting	Height
231205	827533	838645	2.9515
231207	826841	838017	3.1315
8100.001	830539.604	839931.779	unavailable
8056.001	834261.720	835750.350	unavailable
8032.005	834154.362	835760.069	unavailable

Station ID	SPP position error (m)	SPP vertical error (m)	DGNSS position error (m)	DGNSS vertical error (m)
231205	42.99688	71.1945	50.28752	47.9835
231207	26.17929	1.6455	31.37279	24.4495
8100.001	64.70391		64.59643	
8056.001	30.99932		31.11348	
8032.005	44.08571		40.15077	

The data collected reveals a concerning trend regarding the accuracy of the uncorrected GNSS data. The positional error averaging around 50 meters. Such high magnitude of error often suggests that there might be incomplete implementations of the current positioning and DGNSS correction model.

Problems

The most concerning issue during the preliminary survey is the unexpectedly high positional error of the single point positioning. The deviations constantly falling between 30 and 60 meters, the raw data is lower quality than anticipated for modern mobile devices. This excessive noise also makes it difficult to tell whether DGNSS corrections is in fact functioning, as the signal errors may mask the underlying corrections.

From a usability perspective, the current application user interface (UI) does not have the functionality to view the results of both standalone GNSS and assisted with DGNSS. Furthermore, the user cannot directly view the error between the measurements and the known station coordinates. To observe the difference, I edit the source code and rerun the application entirely, which is an inefficient workflow for field testing, and manually do the calculations outside the application.

```
        if (navDataList == null) {
            showDialog( title: "Error",  msg: "Unable to parse nav data csv")
            return
        }
        if (obsDataList == null) {
            showDialog( title: "Error",  msg: "unable to parse obs data csv")
            return
        }

        val obsDataListWithRange = computePseudorange(obsDataList)
        if (stationRtcmMapOptional != null) {
            Log.d( tag: "GNSS",  msg: "Successfully adjusted with DGNSS")
//            adjustPseudorange(obsDataListWithRange, stationRtcmMapOptional)
        }

        showDialog( title: "Success",  msg: "Stored `$baseName` to survey data list")
        surveyDataList.add(SurveyData(baseName, navDataList, obsDataListWithRange))
        createSurveyDataButton()
    }
```

This figure shows the pseudorange adjustment function commented out. Re-running the code is required to see the differences, which is likely not user friendly.

Another issue involves handling the coordinates. The application displays the latitude-longitude-altitude (WGS84) along with the ECEF location. However, the known points provided by LandsD are in Hong Kong 1980 Grid with northing, easting, and height values. To facilitate calculation and input the known points to directly see the accuracy of standalone SPP and DGNSS-assisted measurements, an input field for known coordinates and a toggle for DGNSS is needed.

This screenshot shows the SPP coordinates of 231205 without DGNSS. The UI displays the WGS84 coordinates directly.

Possible Solutions and future works

a. Improve accuracy of standalone GNSS

Firstly, to address the accuracy issues, the standalone SPP must be improved. In the current implementation, the app only retains the latest measurement per PRN, though it simplifies early prototyping and processing, it discarded all previous measurements, thus the redundancy is greatly reduced. In the future works, the least squares adjustments should also include the previous measurements, but a lower weight might be given. Furthermore, I will focus on the implementation of a full Klobuchar model to mitigate ionospheric delays, as well as a tropospheric model, such as the Saastamoinen model, to account for lower atmospheric refraction. Additionally, other subtle error sources, including the earth tidal effect, should be modeled to ensure that the physical environment is represented as accurately as possible in the processing engine.

b. Implement HK1980 Grid transformation

Secondly, the output position should also be transformed into Hong Kong 1980 Grid for easier record keeping and accuracy calculations. This skips the steps of using the web transformation tool by LandsD to convert coordinates.

c. Display both standalone SPP and DGNSS results

As previously mentioned, to improve the user friendliness and for effective demonstration of the accuracies, the app should enable inputting the known coordinates and toggle on and off for DGNSS, or display the accuracies directly with/without DGNSS.

d. Expansion of satellite constellation Support

Currently, the application only handles GPS satellites. As a result, only RTCM type 1 is handled. In the upcoming works it is possible to also handle GLONASS ephemerides, thus decoding RTCM type 31.

```
D Permission: Full
D HTTP Response: ICY 200 OK
D Received DGNSS message type 1 from station 1012!
D Received DGNSS message type 1 from station 1012!
D Received DGNSS message type 1 from station 1012!
W Received RTCM message with unhandled type: 31 from station 1012
D Received DGNSS message type 1 from station 1012!
W Received RTCM message with unhandled type: 31 from station 1012
D Received DGNSS message type 1 from station 1012!
D Received DGNSS message type 1 from station 1012!
D Received DGNSS message type 1 from station 1012!
D Received DGNSS message type 1 from station 1012!
D Received DGNSS message type 1 from station 1012!
D Received DGNSS message type 1 from station 1012!
D Received DGNSS message type 1 from station 1012!
D Received DGNSS message type 1 from station 1012!
W Received RTCM message with unhandled type: 31 from station 1012
D Received DGNSS message type 1 from station 1012!
D Received DGNSS message type 1 from station 1012!
D Received DGNSS message type 1 from station 1012!
```

Receiving DGNSS messages for both GPS and GLONASS, printing to the console log.

Timetable

Week	Task
Week 2 – Week 3	UI and coordinate conversion
Week 3 – Week 6	Complete correction models
Week 6	Field testing and surveys
Week 6 – Week 8	Accuracy analysis
Week 8 – Week 12	Final report writing

Key Dates

25/1/2026 (Week 2)	Progress report
18/3/2026 (Week 9)	Progress presentation
5/4/2026 (Week 11)	Draft report
12/4/2026 (Week 12)	Final report
Week 13 – Week 14	Final presentation (date TBD)

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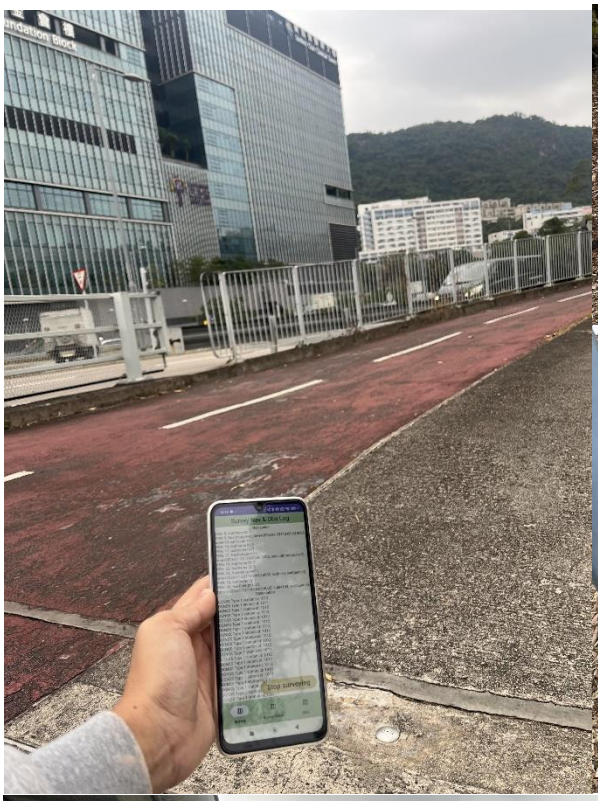
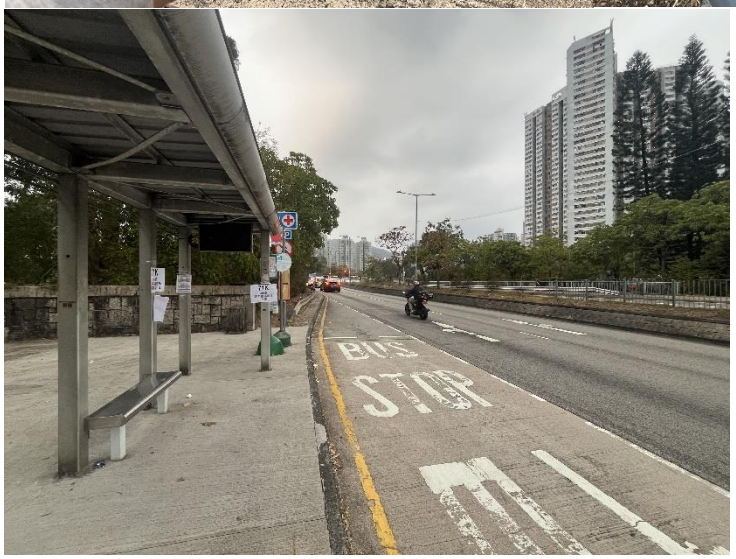
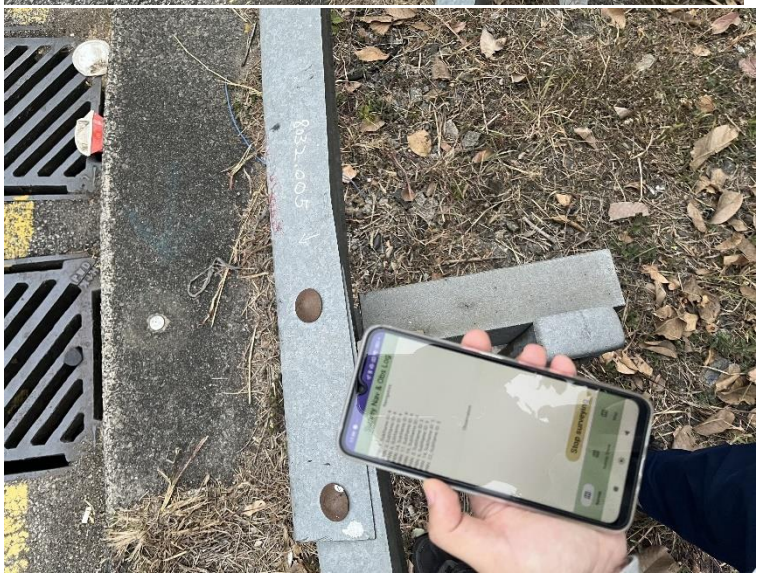
Appendix

Survey photos


stop surveying

16AM


BUS
STOP

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